

Problem Solving in Engineering (IP-3)

Module 2 Workbook: Serenity VR

Systems Mapping & Root Cause

Dissecting Problems using the Iceberg Model and Stakeholder Ecosystems

Zoom in to Your Scope Area

Scope: The anxiety therapy delivery system in urban India - specifically the journey from a patient experiencing anxiety symptoms to receiving (or failing to receive) effective, adaptive, biofeedback-driven treatment.

Identify Events (Visible Issues)

The following visible events, issues, and situations are observed within the system scope:

Main Problem Events:

1. Patients with anxiety disorders are unable to find available mental health professionals within a reasonable timeframe (wait times of 2–8 weeks).
2. Therapy sessions follow fixed protocols (CBT, talk therapy) that do not adapt to the patient's real-time physiological state.
3. There is zero biometric monitoring during therapy - clinicians rely entirely on patient self-reporting, which is subjective and often inaccurate.
4. Patients drop out of therapy after 3–5 sessions due to high costs (Rs.500–3000/session), travel burden, and lack of perceived real-time progress.

Trigger Events (happen just before the problem):

5. A patient experiences an acute anxiety episode or panic attack but cannot access immediate professional help.
6. A student faces exam stress, workplace pressure, or social anxiety but finds no affordable local mental health support.
7. Post-COVID mental health awareness increased, but the infrastructure did not scale proportionally.

Effect Events (happen as a result of the problem):

8. Untreated anxiety compounds into more severe conditions (depression, substance abuse, chronic insomnia).
9. Patients resort to self-medication (over-the-counter sedatives, alcohol) without professional guidance.
10. Workplace productivity drops; absenteeism increases due to unmanaged anxiety disorders.
11. Families of patients face emotional and financial burden from managing untreated mental health conditions.

Indirectly Connected Events:

12. Rise in mental health apps (Calm, Headspace) shows demand, but these apps lack biometric integration, clinical oversight, and adaptive therapy.
13. VR headset prices have dropped significantly (Meta Quest 3 at ~\$500), making immersive therapy technologically feasible for the first time.
14. Local AI models (Llama 3, Mistral) can now run on consumer hardware, enabling personalized guidance without cloud dependency or privacy concerns.

Identify Events & Patterns

Recurring Patterns Identified:

Stakeholder Group Analysis

Stakeholder Group 1: Anxiety Patients (End Users)

Process & Lifecycle from their perspective:

```
Experiences anxiety symptoms (racing heart, shortness of breath, panic)
|
v
Denial / self-management phase (tries breathing exercises, meditation apps)
|
v
Symptoms worsen or become frequent -> decides to seek professional help
|
v
Searches online for psychiatrist/psychologist -> finds limited options
|
v
Books appointment -> waits 2-8 weeks
|
v
Attends first session (30-60 min) -> discusses symptoms verbally
|
v
Receives generic treatment plan (CBT exercises, possible medication)
|
v
```

Returns home -> no monitoring, no feedback between sessions

|

v

Tries to follow exercises on own -> lacks motivation without real-time feedback

|

v

Misses follow-up appointments (cost, time, travel, stigma)

|

v

Symptoms return / worsen -> CYCLE REPEATS or patient gives up entirely

Lifecycle: This cycle repeats every 1–3 months per patient, with each iteration increasing the probability of permanent dropout.

Stakeholder Group 2: Clinicians / Therapists

Process & Lifecycle from their perspective:

Receives appointment booking (already overbooked - 15-30 patients/day)

|

v

Conducts 30-60 min session -> asks patient to self-report symptoms

|

v

Forms assessment based on SUBJECTIVE verbal input only (no physiological data)

|

v

Prescribes treatment plan (CBT, medication, relaxation techniques)

|

v

Schedules next session in 1-2 weeks -> has NO visibility into patient between sessions

|

v

Patient may or may not return for follow-up

|

v

If patient returns: reassesses based on patient's memory of past week (unreliable)

If patient doesn't return: no way to track or intervene

|

v

Clinician moves to next patient -> limited time per patient -> quality suffers

Lifecycle: Daily cycle, repeated for each of the 15–30 patients seen per day. Burnout risk is high.

Stakeholder Group 3: Healthcare Institutions / Hospitals

Process & Lifecycle from their perspective:

Allocates budget for mental health department (typically underfunded)

|

v

Hires limited psychiatrists/psychologists (shortage in market)

|

```

      v
Opens appointment slots -> slots fill up immediately (excess demand)
  |
      v
Long waitlists form -> patient dissatisfaction increases
  |
      v
Patients seek private practitioners or give up entirely
  |
      v
Institution cannot demonstrate treatment outcomes (no longitudinal data)
  |
      v
Difficult to justify increased mental health budget -> funding stays flat
  |
      v
CYCLE REPEATS: understaffed, overbooked, no outcome data

Lifecycle: Annual budget cycle, with the gap widening each year.

```

Identify the System Structure

Going through the process step by step, the following structures enable/constrain the system:

Explore Mental Models

Mental Model Analysis per Stakeholder:

Common Expectations: All stakeholders want better outcomes, reduced cost/effort, and objective measurable data.

Conflicting Priorities: Patients want immediate access vs. Institutions want cost-controlled services. Clinicians want more time per patient vs. Institutions want more patients per clinician.

Shared Assumptions (that may be wrong): "Effective therapy MUST involve a human therapist in real-time." "Biometric monitoring is too expensive." "Scaling therapy = hiring more professionals."

Iceberg Model Summary

```

=====
                        ICEBERG MODEL
=====

```

EVENTS (Tip - What we see):

-
- 85-90% treatment gap for anxiety in India
 - Patients waiting 2-8 weeks for appointments
 - High therapy dropout after 3-5 sessions
 - No real-time biometric data in therapy
 - Rising anxiety prevalence (120%+ since 1990)

PATTERNS (Below surface - What keeps happening):

-
- Consistent year-over-year widening of demand-supply gap
 - Recurring patient dropout cycle every 1-3 months
 - Therapy outcomes plateau due to disconnected sessions
 - Daily anxiety triggers (commute, work) misaligned with weekly therapy

STRUCTURES (Deeper - What allows this):

- Only 0.3 psychiatrists per 100,000 (vs WHO's 3.0)
- No biometric sensor integration in therapy workflows
- Therapy requires physical co-presence (no VR/remote)
- Fixed-protocol CBT not adaptive to real-time state
- High cost structure (Rs.500-3000/session)

MENTAL MODELS (Root - Why the system is this way):

- "Effective therapy must have a human therapist present"
 - "Scaling = hiring more doctors (no tech alternative)"
 - "Mental health stigma discourages seeking help"
-

Review Your Project - Bottlenecks Identified

1. Key Data Bottleneck:

No real-time physiological data flows from patient to clinician. Therapy decisions are based entirely on subjective self-reports. The system has zero objective biometric input during or between sessions.

2. Key Process Bottleneck:

Therapy delivery requires physical co-presence and follows fixed schedules. The current process mandates in-person visits on pre-set appointments, creating a single point of failure.

3. Key Human Bottleneck:

Critical shortage of mental health professionals acts as a hard capacity limit. With only ~0.3 psychiatrists per 100,000 population, no amount of process improvement can address the fundamental supply constraint.

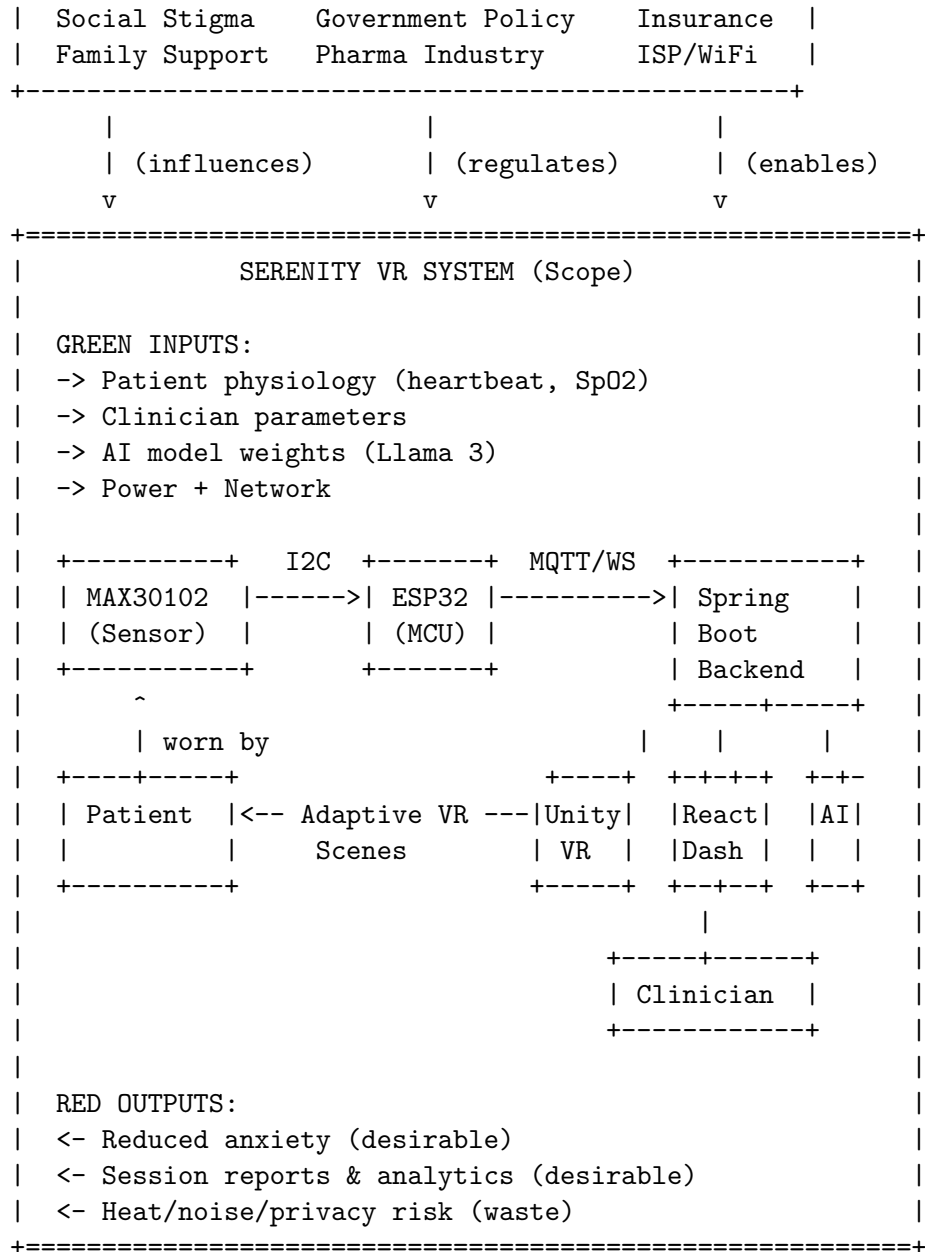
Reviewing "How Might We" from Module 1

- **HMW 1 (Process Bottleneck):** VR-based anxiety therapy at home eliminates the requirement for physical co-presence.
- **HMW 2 (Data Bottleneck):** Local AI and wearable sensors create a continuous biometric data pipeline.
- **HMW 3 (Human Bottleneck):** Continuous biometric tracking with clinician dashboards multiplies clinician reach.

Complete the System Map (Revised from Module 1)

ENVIRONMENT

+-----+



Pattern #	Pattern Description	Frequency	Conditions
P1	Patients delay seeking help due to stigma and lack of awareness	Continuous	Strongest in rural/conservative areas
P2	Long wait times (2–8 weeks) to see a psychiatrist in urban India	Chronic	Worse during exam seasons (Mar-May)
P3	High therapy dropout rates after 3–5 sessions	Recurring per patient	Triggered by cost, travel, lack of progress
P4	Therapy outcomes plateau due to disconnected sessions	Recurring	Every therapy cycle with spaced sessions
P5	Clinicians lack objective physiological data	Every session	Universal across all current methods
P6	Anxiety symptoms peak during morning commutes/work hours	Daily cycle	Urban working professionals and students
P7	Gap between mental health awareness and	Yearly trend	Widening year over year since 2020

Process Step	Underlying Structure
Patient experiences symptoms	Biological/neurological predisposition; environmental stressors (work, academics)
Patient delays seeking help	Social stigma; lack of awareness; cultural beliefs ("anxiety is not a real illness")
Long wait times	Severe shortage: ~0.3 psychiatrists per 100,000 population; geographic concentration
Fixed-protocol therapy	Clinical training models emphasizing standardized CBT; no biometric infrastructure
No monitoring between sessions	Absence of wearable devices in workflows; no data pipeline to clinician
Patient drops out	High cost structure; no tangible progress indicators; no gamification
Clinician lacks objective data	No sensor/biofeedback integration; reliance on subjective patient self-reporting
No scalable delivery mechanism	Therapy requires physical presence; no VR/telemedicine infrastructure
AI not used in therapy	Local AI inference on consumer hardware is a recent development (2023-2024)

Aspect	Anxiety Patient	Clinician / Therapist	Healthcare Institution
Belief	"Therapy should give me instant relief"	"I can only assess what the patient tells me"	"Mental health is not revenue-generating enough"
Expectation	Affordable, convenient, private, effective	More patients seen per day, better tools	Measurable outcomes to justify budget
Assumption	"Technology can't help anxiety" / "VR is for gaming"	"Biometrics are too complex to integrate"	"Scaling requires hiring more doctors"
Priority	Immediate symptom relief; low cost; no stigma	Accurate diagnosis; manageable workload	Cost efficiency; reputation; compliance
Value	Privacy, convenience, tangible progress tracking	Evidence-based tools, reliable data	Scalability, ROI, patient satisfaction